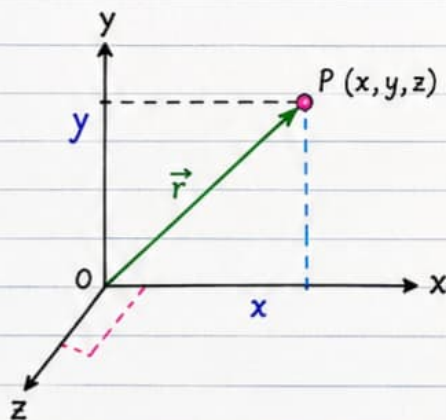


## 1. INTRODUCTION

- When a body moves such that its path is **not** a straight line, the motion is said to be **two dimensional or motion in a plane**.
- The position of a particle in a plane is completely described by **two independent coordinates**.
- Examples** : Projectile motion, circular motion, upward & downward throw of a body, etc.

**Key Point**

In a plane (x-y plane),  $z = 0$   
So, position of P is given by (x, y) only.

## 2. POSITION VECTOR

- The position of a particle P in a plane can be described by its position vector  $\vec{r}$ .
- If the position of P is (x, y), then

$$\vec{r} = x\hat{i} + y\hat{j}$$

(for motion in x-y plane)

**Unit Vectors**

$\hat{i} \rightarrow$  unit vector along +x axis

$\hat{j} \rightarrow$  unit vector along +y axis

## 3. DISPLACEMENT AND VELOCITY

- Displacement** : Change in position vector.

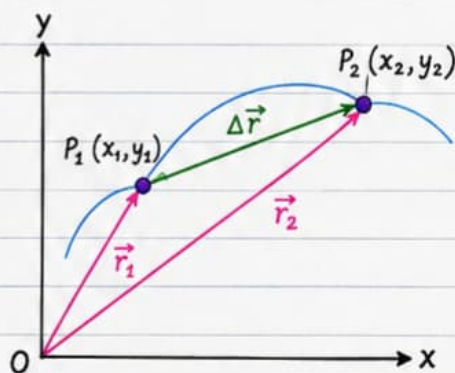
$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

- Velocity** : Rate of change of position vector.

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$$

- Speed** : Magnitude of velocity.

$$v = |\vec{v}| = \sqrt{v_x^2 + v_y^2}$$



Displacement depends only on initial and final position, not on path.

## 4. TYPES OF MOTION IN A PLANE

## (ii) RECTILINEAR MOTION

- When the path of the particle is a **straight line**.
- Example** : Motion of a train on a straight track.



## (ii) CURVILINEAR MOTION

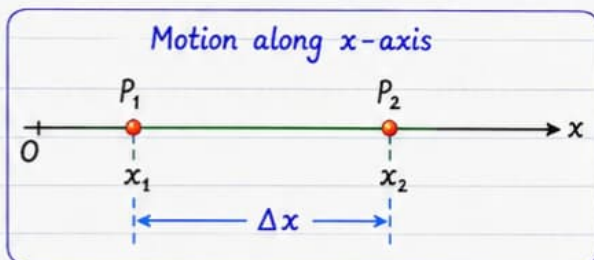
- When the path of the particle is a **curve**.
- Example** : Motion of a thrown ball, planet around the sun.



## 5. RECTILINEAR MOTION

- When a particle moves along a straight line, its motion is completely described by displacement, velocity and acceleration in **one dimension**.

**Example :** A car moving on a straight road, a lift moving up or down, etc.



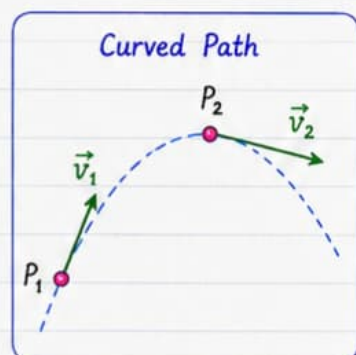
## 6. CURVILINEAR MOTION

- When a particle moves along a curved path, its position, velocity and acceleration must be described in **two dimensions**.

**Example :** Motion of a stone thrown obliquely, motion of a planet, cycled on a circular track, etc.

## Key Idea

In curvilinear motion, both the magnitude and direction of velocity may change.



## 7. PROJECTILE MOTION

- Projectile motion is a special type of motion in a plane.
- An object projected into air with some initial velocity and then moves under the action of gravity alone (neglecting air resistance).
- The path followed is a **parabola**.



## Important Points

- At highest point,  $v_y = 0$  but  $v_x \neq 0$ .
- Time of ascent = Time of descent.
- Range (R) is the horizontal distance travelled.
- Maximum height ( $H_{\max}$ ) is the greatest vertical distance.

## Mathematical Expressions (For level projection)

$$\text{Time of flight (T)} = \frac{2u \sin \theta}{g}$$

$$\text{Range (R)} = \frac{u^2 \sin 2\theta}{g}$$

$$\text{Maximum height (H}_{\max}) = \frac{u^2 \sin^2 \theta}{2g}$$

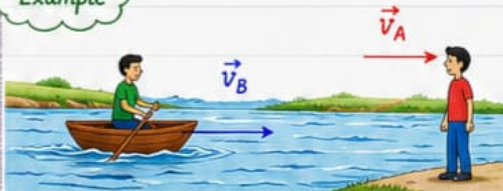
$u$  = Initial speed  
 $\theta$  = Angle of projection  
 $g$  = Acceleration due to gravity ( $9.8 \text{ m/s}^2$ )

## 8. RELATIVE VELOCITY

- If two objects are moving, the velocity of one object as observed from another is called **relative velocity**.
- If  $\vec{v}_A$  is velocity of A w.r.t. ground and  $\vec{v}_B$  is velocity of B w.r.t. ground, then velocity of A w.r.t. B is

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$$

## Example



$\vec{v}_A$  = Velocity of man w.r.t. ground  
 $\vec{v}_B$  = Velocity of boat w.r.t. water  
 $\vec{v}_{A/B}$  = Velocity of man w.r.t. boat

Relative velocity is important in problems involving boats, rivers, wind, moving platforms etc.

## 9. ADDITION AND SUBTRACTION OF VECTORS

- In two dimensions, vectors are added or subtracted by **components** or by **graphical methods**.

## (A) By Components (Analytical Method)

Let  $\vec{A} = A_x \hat{i} + A_y \hat{j}$   
 $\vec{B} = B_x \hat{i} + B_y \hat{j}$

Then,

$$\vec{A} + \vec{B} = (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j}$$

$$\vec{A} - \vec{B} = (A_x - B_x) \hat{i} + (A_y - B_y) \hat{j}$$

- Magnitude of resultant:

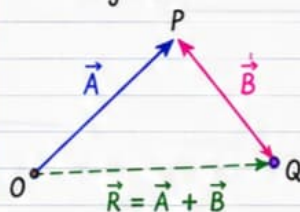
$$R = |\vec{A} + \vec{B}| = \sqrt{R_x^2 + R_y^2}$$

- Direction of resultant ( $\theta$ ):

$$\tan \theta = \frac{R_y}{R_x}$$

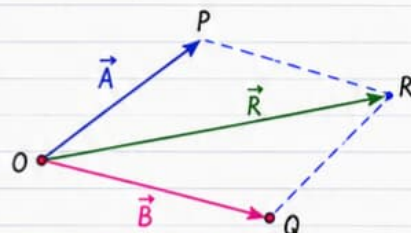
## (B) By Graphical Method (Triangle/Parallelogram Law)

- Triangle Law:



Resultant  $\vec{R}$  is drawn from tail of first vector to head of second vector.

- Parallelogram Law:



$\vec{R}$  is the diagonal of the parallelogram formed by  $\vec{A}$  and  $\vec{B}$ .

## Example:

Two vectors  $\vec{A} = 3\hat{i} + 4\hat{j}$  and  $\vec{B} = 1\hat{i} - 2\hat{j}$  are given.

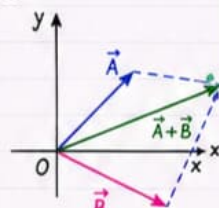
Find  $\vec{A} + \vec{B}$ ,  $\vec{A} - \vec{B}$  and magnitude of  $\vec{A} + \vec{B}$ .

## Solution:

$$\vec{A} + \vec{B} = (3+1)\hat{i} + (4-2)\hat{j} = 4\hat{i} + 2\hat{j}$$

$$\vec{A} - \vec{B} = (3-1)\hat{i} + (4+2)\hat{j} = 2\hat{i} + 6\hat{j}$$

$$|\vec{A} + \vec{B}| = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$$



## 10. CONDITIONS FOR PROJECTILE MOTION

- A body is said to be a projectile if after projection, it moves **only under the action of gravity**.
- Conditions:
  - The body is projected into air with some initial velocity.
  - Air resistance is neglected.
  - The only force acting on the body after projection is gravity.

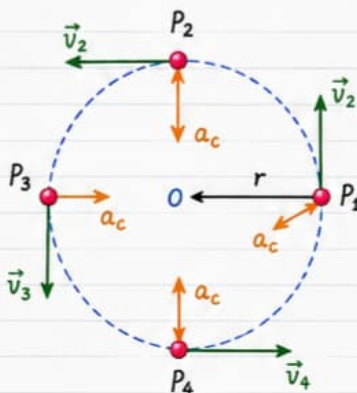


Path is parabolic in the absence of air resistance.

## 11. UNIFORM CIRCULAR MOTION

- When a particle moves along a circular path with **constant speed**, it is said to be in **uniform circular motion**.
- The direction of velocity changes continuously, hence there is acceleration **towards the centre**.
- Centripetal Acceleration:

$$a_c = \frac{v^2}{r} = \omega^2 r$$



## Important Relations

- Angular speed,  $\omega = \frac{v}{r}$
- Time period,  $T = \frac{2\pi r}{v} = \frac{2\pi}{\omega}$
- Frequency,  $f = \frac{1}{T} = \frac{\omega}{2\pi}$

$r$  = Radius of circle  
 $v$  = Linear speed  
 $\omega$  = Angular speed  
 $T$  = Time period

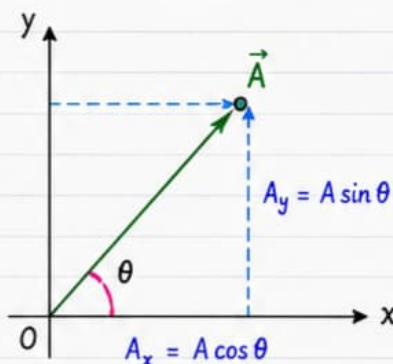
★ Motion in a plane is the combination of ideas from straight line and circular motion. ♥

## 12. RESOLUTION OF A VECTOR

- Any vector  $\vec{A}$  can be resolved into two component vectors along two mutually perpendicular axes.
- If  $\vec{A}$  makes an angle  $\theta$  with +x axis,

$$A_x = A \cos \theta \quad (\text{along } +x \text{ axis})$$

$$A_y = A \sin \theta \quad (\text{along } +y \text{ axis})$$



## Note

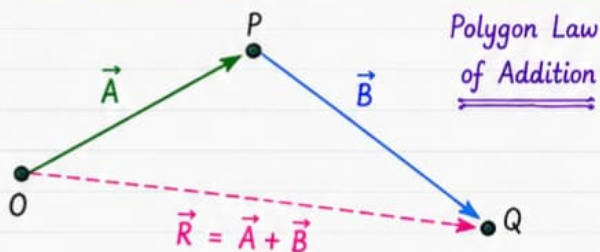
The components completely determine a vector and can be used to add or subtract vectors easily.

## 13. ADDITION OF VECTORS (GRAPHICAL METHOD)

- The sum of two vectors can be found by polygon law of addition.

## Steps :

- Draw first vector  $\vec{A}$  to some scale.
- From the head of  $\vec{A}$ , draw the second vector  $\vec{B}$  to the same scale.
- The resultant  $\vec{R} = \vec{A} + \vec{B}$  is the vector from tail of  $\vec{A}$  to head of  $\vec{B}$ .



## Polygon Law of Addition

For two vectors :

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

where  $\theta$  is the angle between  $\vec{A}$  and  $\vec{B}$ .

## 14. APPLICATIONS

- (i) Projectile motion  $\rightarrow$  Path of a cannon ball, ball thrown obliquely.
- (ii) Uniform circular motion  $\rightarrow$  Motion of satellites, planets, wheels.
- (iii) Relative velocity  $\rightarrow$  Boats in a river, moving walkways, wind problems.



- ★ Motion in a plane involves two-dimensional motion described using vectors.
- ★ Displacement, velocity and acceleration are vector quantities.
- ★ Rectilinear motion  $\rightarrow$  along a straight line (1-D). Curvilinear motion  $\rightarrow$  along a curve (2-D).
- ★ Projectile motion  $\rightarrow$  motion under gravity along a parabolic path.
- ★ Uniform circular motion  $\rightarrow$  constant speed in a circular path, with centripetal acceleration.
- ★ Vectors can be added/subtracted using components or graphical methods.

## IMPORTANT FORMULAS

$$\vec{r} = x \hat{i} + y \hat{j}$$

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

## KEY TAKEAWAYS

- ✓ Always draw a neat diagram.
- ✓ Choose convenient axes for resolving vectors.
- ✓ For level problems, use projectile formulas.
- ✓ Practice vector addition both analytically (components) and graphically.

